

A hybrid deep learning-Muskingum framework for enhanced runoff prediction: Model coupling and hydrological process integration

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ABSTRACT

Study region: The upper reaches of the Hanjiang River, China

Study focus: To enhance the accuracy and physical consistency of reservoir inflow forecasting, this study proposes a hybrid modeling framework that couples an enhanced Muskingum model with a bidirectional long short-term memory (BiLSTM) network. The Muskingum model was restructured via differentiable programming to allow dynamic calibration of physical parameters across river sub-reaches. This physics-based layer was embedded within the BiLSTM network to learn the relationship between meteorological forcing inputs and runoff dynamics. Bayesian Optimization (BO) was adopted to co-optimize the Muskingum parameters and neural network hyper-parameters, mitigating error propagation and thus enhancing predictive robustness.

New hydrological insights for the region: The proposed framework was evaluated on a reach of the upper Hanjiang River between Ankang and Danjiangkou Reservoirs. Results showed that model performance initially improved with finer segmentation, peaking with a four-segment configuration, after which performance declined—likely due to over-parameterization. The optimal four-segment hybrid model achieved a Nash–Sutcliffe efficiency (NSE) of 0.94 during the test period, representing a 4.4 % improvement over both the pure BiLSTM model and the one-way coupled model. In addition, it achieved a Kling–Gupta efficiency (KGE) of 0.95 and a Root Mean Square Error (RMSE) of 598 m³ /s, exhibiting more stable predictive behavior. This further demonstrates the framework's capability for accurate runoff characterization and high-precision inflow forecasting in complex reservoir systems.

1. Introduction

Accurate prediction of reservoir inflow runoff is critical for the effective management of water resources, as it plays a significant role in ensuring the operational safety of reservoir regulation decisions (Liu et al., 2023). However, the impacts of climate change and anthropogenic activities have introduced considerable nonlinearity and non-stationarity into runoff processes, thereby necessitating more stringent accuracy requirements for forecasting models (Jing et al., 2023; Moosavi et al., 2022). The inflow to reservoirs is influenced not only by the direct discharge from upstream reservoirs but also by a combination of various factors, including watershed

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precipitation and meteorological conditions. This multifaceted driving effect further complicates the system (Song et al., 2022; Twinomuhangi et al., 2025). As a result, the development of model architectures that can effectively incorporate complex physical processes and nonlinear dynamics, along with the scientific calibration of parameters, has emerged as a primary challenge in contemporary runoff forecasting (Feigl et al., 2022).

While traditional hydrological approaches, with their transparent physical basis, continue to underpin flood simulation and reservoir management practices (Aletaha et al., 2024; Moradi et al., 2023), their reliance on static assumptions and simplified process representations limits their capacity to model nonlinearities and temporal variability in modern hydroclimatic regimes. In recent years, deep learning has demonstrated strong pattern-fitting capabilities in hydrological forecasting, particularly when supported by large datasets (Granata et al., 2024). This has facilitated the identification of latent patterns and the modeling of intricate systems (Rebello et al., 2025; Zhang et al., 2025). However, the omission of physical mechanisms in exclusively data-driven models may result in diminished robustness and interpretability when faced with the non-stationarity and complexity inherent to extreme event prediction. (Tripathy and Mishra, 2024; Xie et al., 2021). In response to this trend, the advancement of coupled models that integrate physical constraints with the predictive capabilities of deep learning has emerged as a critical approach to addressing existing challenges in the forecasting of complex watershed runoff. (Li et al., 2024).

In examining the dynamics of inflow and outflow between upstream and downstream reservoirs, the classical Muskingum method, as introduced by McCarthy (1938), simplifies the flow motion equation through the assumption of a linear reservoir. This approach establishes a linear recursive relationship between storage and discharge, which has led to its widespread application in engineering practice due to its succinct structure and well-defined physical parameters. However, the fixed-parameter nature of this method limits its effectiveness in highly nonstationary flow conditions (Perumal and Price, 2013). To overcome the shortcomings of conventional runoff routing models, researchers have progressively developed enhanced Muskingum models with variable parameters. For instance, Niazkar and Afzali. (2017) proposed a nonlinear variable-parameter Muskingum model that utilizes an adaptive differential evolution algorithm, segmenting the entire flow process into three distinct sub-periods and optimizing parameters independently for each segment, thereby significantly reducing simulation errors. More recently, Haiati et al. (2025) presented a Muskingum runoff routing method that combines data clustering with the vulture optimization algorithm. This innovative approach involves grouping runoff data into clusters and employing global optimization techniques, which markedly enhances the accuracy of parameter estimation.

Recent advancements in research have focused on integrating physical equations directly into neural network architectures, facilitating the concurrent optimization of both physical parameters and network hyperparameters (Solanki et al., 2025). Notably, a distributed physics-deep learning hybrid model has demonstrated significant efficacy in the Amazon Basin (Wang et al., 2024). Li et al. (2024) introduced a hybrid framework wherein outputs from physical models, such as the Hydrologic Engineering Center's Hydrologic Modeling System (HEC-HMS), are utilized as inputs for Long Short-Term Memory (LSTM) networks. By training this model with a combination of historical observations and simulation data, they achieved lower prediction errors compared to either physics-based or purely data-driven models in isolation. However, despite these advancements in hybrid physical-machine learning modeling, existing methodologies continue to face challenges in adequately capturing the dynamic interactions of runoff processes, particularly the ongoing exchanges from upstream reservoir releases to downstream inflows. Most current hybrid modeling frameworks employ a sequential architecture, wherein outputs from physical models are simply treated as inputs for machine learning components, lacking intrinsic integration between the two (Xu et al., 2024). Consequently, the calibration process remains disjointed, as physical parameters and neural network hyperparameters are optimized independently, without a coordinated approach. Although hybrid modeling has made significant strides, research addressing the joint calibration of physical and neural components is still limited (Polyakov and Stepanova, 2023).

This study presents an enhanced Muskingum model by reformulating its river-reach flow equations using differentiable programming, which allows for adaptive parameter adjustment across channel segments. The physically-based Muskingum module is integrated into a BiLSTM network, utilizing Global Land Evaporation Amsterdam Model (GLEAM) evapotranspiration and Multi-Source Weighted-Ensemble Precipitation Version 2 (MSWEP-V2) precipitation as inputs. Meteorological features extracted by the BiLSTM, along with inflow estimates derived from the Muskingum model, are dynamically coupled in subsequent layers, establishing bidirectional feedback between storage-discharge equations and hidden-state vectors. Furthermore, a Bayesian optimization framework was developed to integrate the traditionally separate calibration processes for hydrological and neural network parameters, effectively reducing error accumulation during optimization.

2. Materials and methods

2.1. Study area

The Hanjiang River represents the largest tributary of the Yangtze River and is distinguished by significant intra-annual flow variability within the Yangtze River Basin. This research concentrates on the uncontrolled segment of the Hanjiang River's mainstem situated between the Ankang Reservoir and the Danjiangkou Reservoir. The Danjiangkou Reservoir, located across Hubei and Henan provinces, is equipped with multi-year regulation capabilities and receives inflows from several upstream reservoirs. Key hydraulic infrastructures within the catchment area governed by Danjiangkou include the Ankang Reservoir in Shaanxi Province and the Huanglongtan Reservoir in Hubei Province. The study area was defined through GIS-based hydrological analysis, deliberately excluding the contributing catchments of the Ankang and Huanglongtan reservoirs. The selected reach extends approximately 260 km along the mainstem of the Hanjiang River and encompasses a controlled drainage area of roughly 42,000 km², which constitutes 43 % of the upper Hanjiang River basin. The region's topography is characterized by a general decline in elevation from the northwest to the

southeast. Furthermore, the basin experiences significant rainfall and exhibits spatially concentrated precipitation patterns during the flood season. An overview of the topography and reservoir locations within the study area is illustrated in Fig. 1.

2.2. Data

The flow data utilized in this study were sourced from the Hydrological Bureau, Changjiang Water Resources Commission, encompassing daily inflow and outflow records for the Ankang, Huanglongtan, and Danjiangkou reservoirs from 2013 to 2023. Precipitation data were obtained from the MSWEP-V2 dataset, the world's first high-resolution, long-term global precipitation dataset developed by Princeton University (Beck et al., 2019). Evaporation data were derived from the GLEAM dataset, a global evapotranspiration product based on physical models and remote sensing observations, developed by the University of Amsterdam (Miralles et al., 2025). Fig. 2 illustrates the spatial distribution of key hydrological and geographic features within the study area.

Precipitation and evapotranspiration datasets were spatially extracted for the study watershed basin by clipping them to the vector boundary illustrated in Fig. 1. These datasets provided essential hydrological variables for a comprehensive analysis of in-watershed processes. All data were divided into training, validation, and testing subsets in a 6:2:2 ratio, respectively, with a one-day prediction horizon. The temporal allocation for each period is detailed in Table 1.

2.3. Methods

2.3.1. Enhanced Muskingum routing module

The Muskingum method, proposed by McCarthy in 1938, is a widely used technique for channel flood routing in hydrology. Its storage equation is expressed as:

$$W = K[xI + (1 - x)Q] = KQ' \quad (1)$$

where K represents the storage constant, equal to the travel time of the reach under steady-state flow conditions, and x is the flow weighting coefficient, which mainly relates to the degree of wave attenuation and distortion of the flood hydrograph.

In practical applications, once the parameters K and x are established for a specific river reach, the Muskingum routing equations can be utilized in conjunction with the inflow hydrograph at the upstream section and the initial outflow at the downstream section to calculate the outflow hydrograph at the downstream section. For extensive river reaches, the reach is segmented into multiple sub-

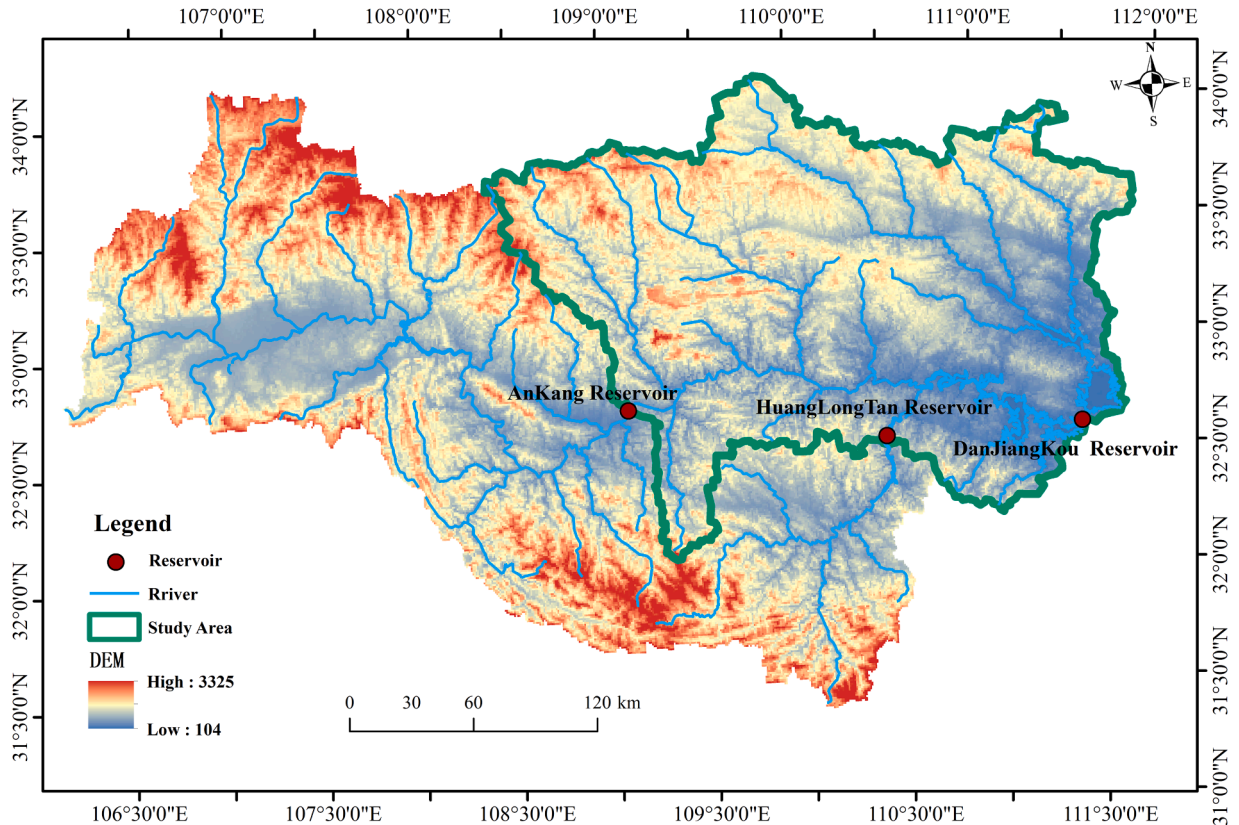


Fig. 1. Schematic map of the upper Hanjiang River basin.

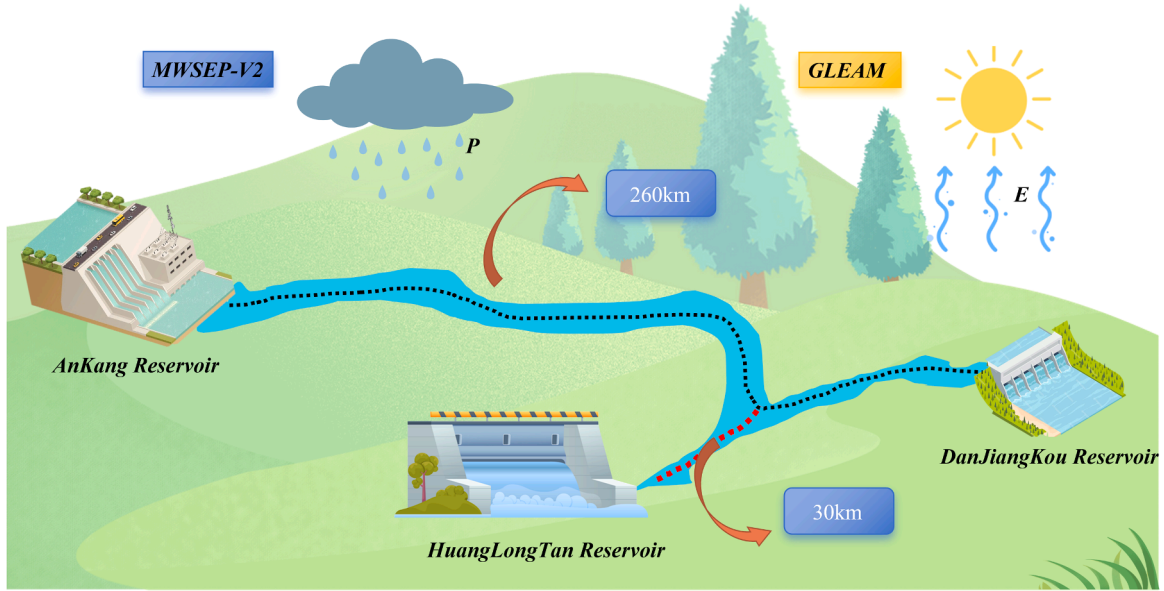


Fig. 2. Relative locations of Ankang, Huanglongtan, and Danjiangkou reservoirs.

Table 1
Dataset partitioning for model development and evaluation.

Period	Duration	Resolution
Training	2013.06–2019.06	daily
Validation	2019.07–2021.06	daily
Testing	2021.07–2023.06	daily

reaches, and the Muskingum method is implemented in a continuous segmented continuous routing approach. The outflow Q from the n th sub-reach serves as the inflow I to the $(n + 1)$ th sub-reach. This recursive relationship ultimately yields the Muskingum confluence hydrograph at the downstream section of the entire river reach.

The nonlinear fluctuations of flow in relation to time and river length, particularly in the context of intricate hydrological features, pose significant challenges to the conventional linear conditioning assumptions, thereby revealing certain limitations (Bindas et al., 2024). This study introduces an advanced methodology that discards the reliance on fixed Muskingum parameters, opting instead for a dynamic calibration of the parameters for each sub-reach through the integration of an optimization algorithm with a neural network. Consequently, the parameters for each short sub-reach are not uniformly applied; rather, they are modified in accordance with the specific flow characteristics of the sub-reach and the spatiotemporal dynamics of the flow. The flow routing equations pertinent to the variable-parameter segmented Muskingum continuous routing algorithm are delineated as follows:

$$Q_{n,j} = C_{0,n}I_{n,j} + C_{1,n}I_{n,j-1} + C_{2,n}Q_{n,j-1} \quad (2)$$

In this context, the segmental flow routing coefficients $C_{0,n}$, $C_{1,n}$, $C_{2,n}$ are determined based on the parameters of the n th sub-reach, and are articulated expressed as follows:

$$\begin{aligned} C_{0,n} &= \frac{(-K_n x_n + 0.5 \Delta t_n)}{(K_n - K_n x_n + 0.5 \Delta t_n)} \\ C_{1,n} &= \frac{(K_n x_n + 0.5 \Delta t_n)}{(K_n - K_n x_n + 0.5 \Delta t_n)} \\ C_{2,n} &= \frac{(K_n - K_n x_n - 0.5 \Delta t_n)}{(K_n - K_n x_n + 0.5 \Delta t_n)} \end{aligned} \quad (3)$$

In Eqs. (2) and (3), $Q_{n,j-1}$, $Q_{n,j}$, $C_{0,n}$, $C_{1,n}$, $C_{2,n}$, $I_{n,j-1}$, $I_{n,j}$, Δt_n all variables represent parameters associated with the n th sub-reach. The enhanced segmented Muskingum continuous routing algorithm allows for the assignment of unique parameters to each short sub-reach during the routing process. This approach facilitates a more precise representation of the intricate hydrological characteristics inherent to each segment, addresses the constraints associated with uniform parameter configurations in conventional methodologies, and enhances both the adaptability and accuracy of flood routing.

2.3.2. Integration of the revised Muskingum model with BiLSTM

In hydrological forecasting, the integration of physical models with machine learning techniques has emerged as a powerful paradigm for enhancing predictive accuracy. Machine learning can self-adjust based on data, while physical models offer a stronger theoretical foundation and greater interpretability (Zhang et al., 2025). In certain instances, the theoretical foundations of physical models may exhibit limitations; nevertheless, with an adequate amount of observational data, these processes can be effectively simulated through a machine learning framework (Bhasme et al., 2021). This research builds upon the improved integration of physical routing and data-driven methodologies by developing a reservoir inflow forecasting model that incorporates a Muskingum routing layer within a BiLSTM network. Specifically, the input variables are first processed through the Muskingum layer to produce a preliminary runoff estimate Q' . This estimate, in conjunction with the original input variables, is subsequently input into the subsequent BiLSTM layers. This process allows the BiLSTM to extract finer temporal features, thereby improving prediction accuracy. The final forecasted discharge is denoted as Q . Within this framework, the differentiable Muskingum module is implemented as a computational layer that reformulates the routing equations to enable automatic differentiation of the parameters K and x . This configuration allows the loss gradients to be backpropagated through both the BiLSTM network and the Muskingum equations, ensuring end-to-end joint optimization of physical and neural components. To further enhance the model's performance, BO is employed to jointly calibrate the segmented Muskingum parameters and the neural network's structural hyperparameters, efficiently exploring the parameter space to identify optimal combinations and thereby significantly enhancing the model's overall performance on both the training and validation datasets. The comprehensive architecture of the model is depicted in Fig. 3.

To improve the reproducibility and clarity of the optimization process, the workflow of the Bayesian Optimization (BO) procedure is illustrated in Fig. 4. BO provides efficient and reliable global search performance in hydrological and environmental model calibration (Zhu et al., 2024; Sun et al., 2024). In this study, BO was chosen because it can efficiently search for the global optimum of complex, non-convex, and computationally expensive objective functions with a limited number of model evaluations. Compared with heuristic or grid-based algorithms (e.g., GA, PSO, or grid search), BO constructs a probabilistic surrogate model that balances exploration and exploitation, thereby improving both optimization efficiency and result stability. This makes it particularly suitable for the physically-machine-learning coupled framework proposed herein, where each model evaluation involves both hydrological simulation and deep-learning training.

The BO algorithm was executed for 50 iterations using a Gaussian Process (GP) surrogate model and the Expected Improvement (EI) acquisition function. The search space was constrained by the physically meaningful ranges of Muskingum parameters and BiLSTM hyperparameters, and convergence was achieved when the maximum EI value fell below 1×10^{-3} .

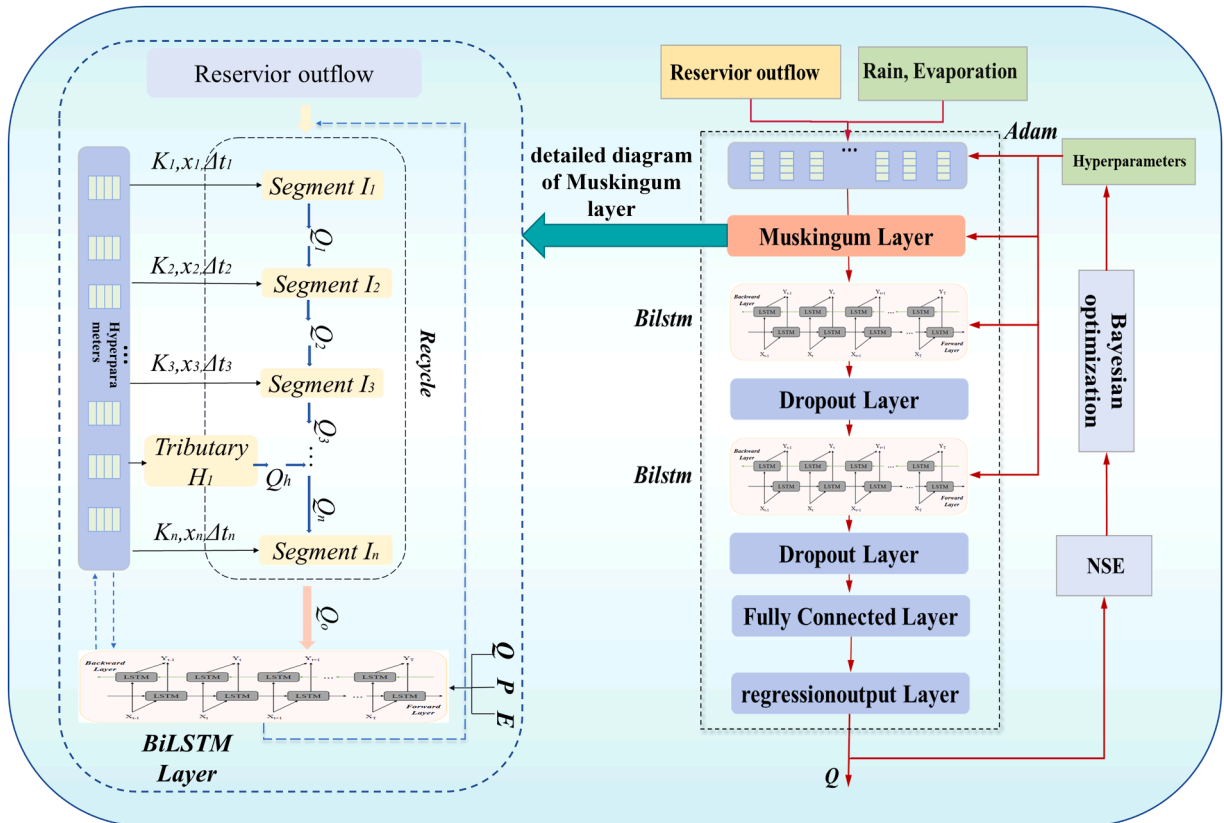


Fig. 3. Coupled Muskingum-Neural network architecture.

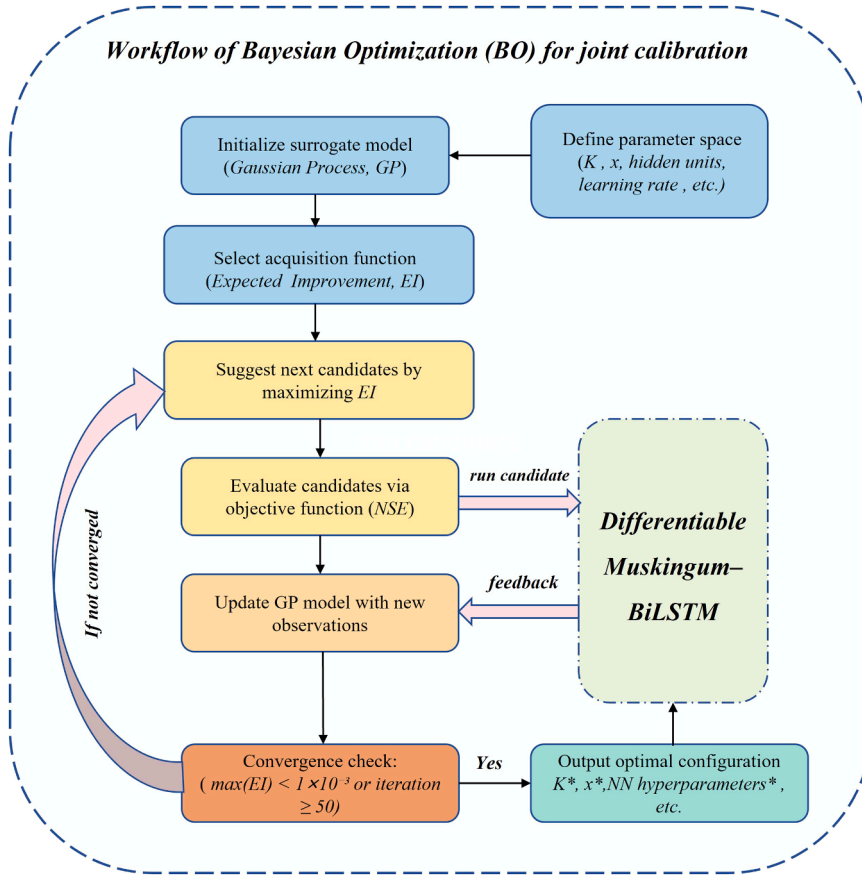


Fig. 4. Workflow of BO for joint calibration.

When applying BO for parameter calibration in practice, appropriate lower and upper bounds should be set for each parameter to be estimated according to the specifics of the problem. The NSE between simulated and observed runoff is used as the objective function for BO, and the selected hyperparameters are optimized accordingly. Additionally, an L_2 regularization term is also incorporated to manage the overall complexity of the model (Cao et al., 2024). All experiments were conducted on the MATLAB 2024 platform, and all schemes were implemented on a workstation equipped with dual NVIDIA RTX 4090D GPUs to ensure computational efficiency and reproducibility. The ranges of the parameters to be optimized are presented in Table 2.

3. Results and discussion

3.1. Input variable screening

In light of the 1-day forecasting horizon established in this research, a versatile, lag-aware, and segment-based simulation framework was devised to more effectively capture the time-dependent effects of upstream reservoirs on the inflow to the Danjiangkou Reservoir. Specifically, this framework considers the operational coordination among reservoirs as well as the spatial distances between critical hydraulic nodes. The maximum lag for the outflow from the Ankang Reservoir was determined to be 3 days, reflecting the approximately 260 km river reach to Danjiangkou. For the Huanglongtan Reservoir, which has a shorter reach of approximately

Table 2

Value ranges of parameters to be optimized.

Parameter Name	Description	Type	Range
<i>InitialLearnRate</i>	Initial learning rate for network training	float	[0.001,0.1]
<i>NumUnits</i>	Number of hidden neurons in each LSTM layer	integer	[1150]
L_2	Weight of the L_2 penalty term in the loss	float	$[10^{-10},0.01]$
<i>MaxEpochs</i>	Maximum number of training epochs or iterations	integer	[1150]
K_n	Muskingum storage constant for the n th segment	float	[1,30]
x_n	Muskingum weighting factor for the n th segment	float	[0,0.45]

40 km, a maximum lag of 1 day was implemented. Consistent with this approach, a 3-day lag was also applied to the precipitation and evaporation processes within the Ankang–Danjiangkou interval basin. To further ascertain the optimal set of input variables, two widely recognized statistical methodologies were utilized. First, Pearson correlation analysis was performed to evaluate the linear relationship between each lagged input and the inflow to the Danjiangkou Reservoir; a higher absolute value of the correlation coefficient (r) signifies a stronger linear association. Subsequently, multiple linear regression analysis was conducted to assess the cumulative effects of all lagged inputs. The contribution of each input variable was quantified through its regression coefficient, while the overall significance of the regression model was evaluated using the F-statistic and the corresponding p-values (Cao et al., 2024; Wan Jaafar et al., 2011).

In the context of variance analysis for the regression model, the F-statistic is used to evaluate the overall significance of the model. The corresponding p-value quantifies the probability of observing such an F-statistic under the null hypothesis, where a smaller p-value indicates stronger evidence that the regression model provides a significantly better fit. In this investigation, the calculated F-value was 2460.42 accompanied by a corresponding p-value of less than 0.0001, which indicates a highly significant regression relationship. For each independent variable, the regression coefficient denotes both the direction and magnitude of its effect on reservoir inflow, while the standard error reflects the precision of the coefficient estimate. The t-value, derived from the ratio of the coefficient to its standard error, is employed to assess the statistical significance of the variable; a larger absolute t-value indicates a stronger association. The corresponding p-value ($\text{Prob} > |t|$) assesses the individual significance of the variable, with smaller values suggesting greater statistical significance (Wang et al., 2025). It is noteworthy that the Pearson correlation coefficients between evaporation and reservoir inflow consistently hovered around 0.1, indicating a limited direct influence and a relatively minor contribution to the model. In the multiple linear regression analysis, although the p-value of $E(t-2)$ was marginally higher than those of other lagged variables, its relatively large absolute t-value suggested a stronger statistical significance. Furthermore, given the inherent coupling between precipitation and evaporation within atmospheric circulation processes, a synchronous lag structure was deemed appropriate. As a result, both precipitation and evaporation were incorporated into the final model with a two-day lag in the final model. Based on the evaluation metrics (Table 3) and the Pearson correlation analysis (Fig. 5), the optimal combination of input variables has been identified as $AK_O(t-1)$, $HLT_O(t-1)$, $P(t-2)$, and $E(t-2)$.

3.2. Optimal river segmentation configuration

In order to account for the length of the river reach and the inflows from tributaries, this study examined the simulation efficacy of segmenting the reach from the Ankang Reservoir to the Danjiangkou Reservoir into 2–5 distinct segments, followed by a comparative analysis of the outcomes. The simulation methodology is structured as follows: the primary river channel is sequentially divided based on the predetermined number of segments, with runoff routing executed for each segment individually. The outflow from each segment is utilized as the inflow for the subsequent downstream segment, thereby establishing a recursive propagation framework. Upon reaching the confluence at the Huanglongtan Reservoir, the release flow from this reservoir is combined with the main stream, and the routing process continues downstream until it arrives at the control cross-section of the Danjiangkou Reservoir. As illustrated in Figs. 6 and 7, the results obtained from various segmentation strategies generally reflect seasonal flow variations effectively; however, notable discrepancies arise during peak flood events. An increase in the number of segments from 2 to 5 resulted in a "rise-then-stabilize" trend in the model's accuracy regarding the reproduction of flood peaks and the capture of time-lag responses. Among the configurations tested, the four-segment approach exhibited the most favorable overall performance in simulating the magnitude, timing, and recession of flood peaks. It achieved optimal results across all three evaluation metrics: a NSE of 0.94, a KGE of 0.95, and a RMSE reduced to $598 \text{ m}^3/\text{s}$. These results indicate that this configuration strikes a balance between computational complexity and predictive accuracy, effectively characterizing the channel's time-lag response and the effects of tributary inflow for high-precision runoff routing.

Table 3
Analysis of variance for the multiple linear regression model.

variable type	Source Model	Sum of squares 7.49	Mean square 6.2	F value 2509.79	Prob>F < 0.0001
	variables	Coefficient	Standard Error	t value	Prob> t
Reservoir factor	$AK_O(t-3)$	-42.6073	16.3076	-2.6100	0.7465
	$AK_O(t-2)$	0.0062	0.0193	0.3200	5.1933×10^{-28}
	$AK_O(t-1)$	0.2928	0.0265	11.0600	0.0000
	$HLT_O(t-3)$	1.0754	0.0199	54.0000	2.8054×10^{-4}
	$HLT_O(t-2)$	0.2631	0.0724	3.6400	0.1119
meteorological factor	$HLT_O(t-1)$	0.1364	0.0858	1.5900	1.5523×10^{-65}
	$P(t-3)$	1.2637	0.0724	17.4500	7.9386×10^{-34}
	$P(t-2)$	24.6584	2.0133	12.2500	6.4333×10^{-103}
	$P(t-1)$	46.2293	2.0778	22.2500	0.1102
	$E(t-3)$	3.1884	1.9954	1.6000	1.6412×10^{-3}
	$E(t-2)$	-80.2935	14.1889	-5.6600	0.0022
	$E(t-1)$	47.5741	15.5311	3.0600	0.0012

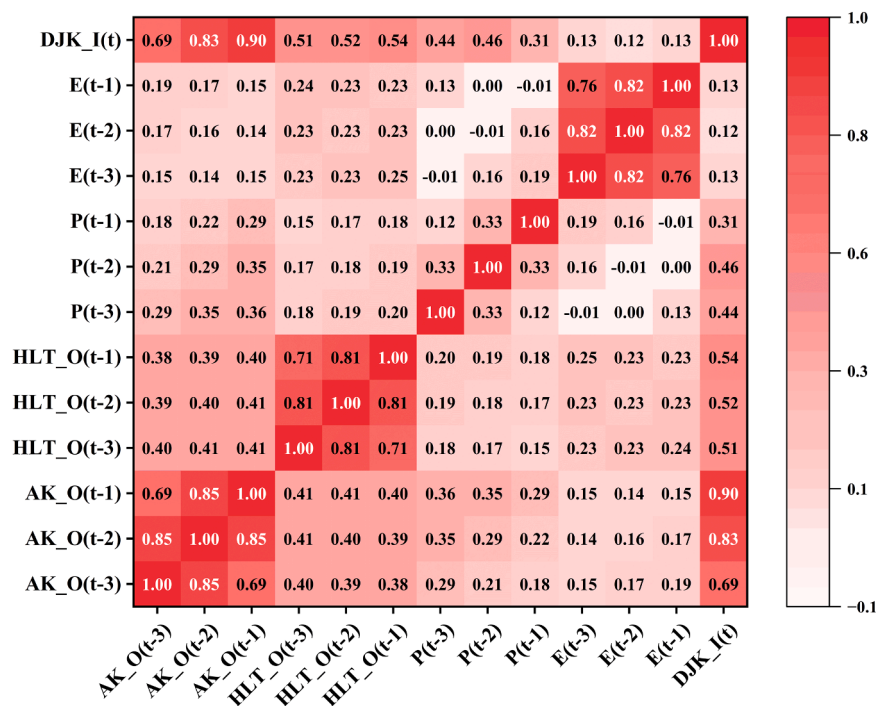


Fig. 5. Pearson correlation coefficients under different time lags.

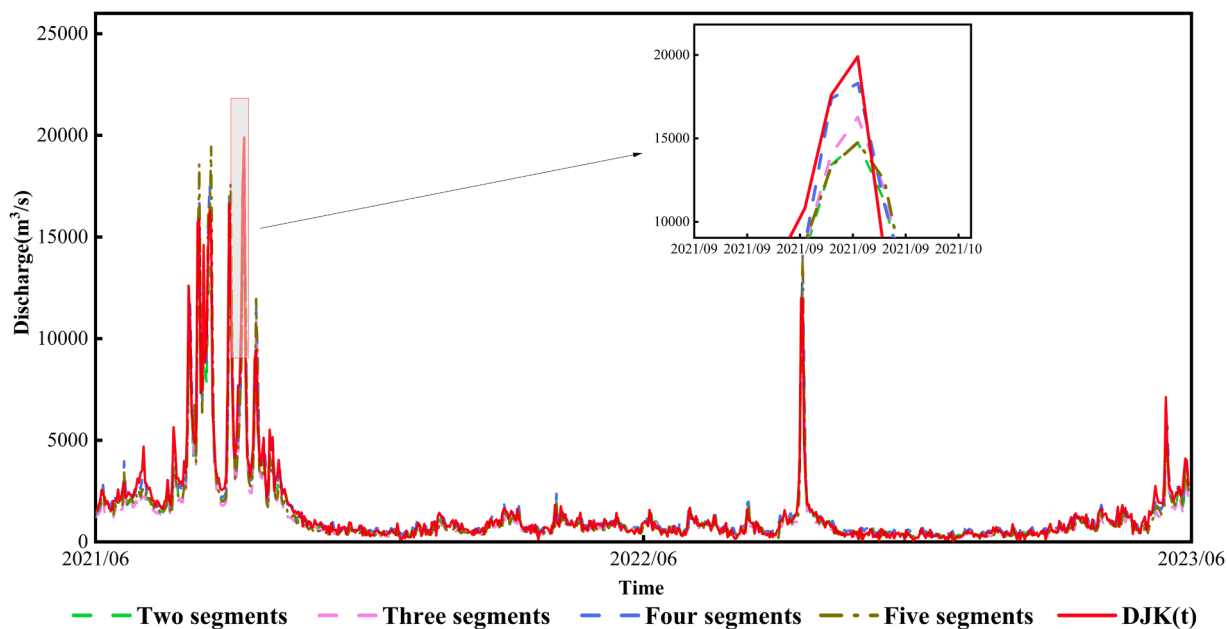


Fig. 6. Comparison of simulated runoff processes under different segments.

3.3. Comparative modeling scenarios and performance evaluation

To further examine the influence of physical constraints, we designed a one-way physical coupling experiment based on the original BO-BiLSTM framework. In this experiment, the discharge simulated by the Bayesian-optimized Muskingum model was introduced into the BO-BiLSTM as an additional input. This coupling strategy is not embedded within the neural network but is applied externally at the framework level, thereby representing a form of shallow physical integration. In contrast, the deep coupled hybrid model proposed in this study embeds the physical routing process directly within the network architecture, allowing the Muskingum parameters and

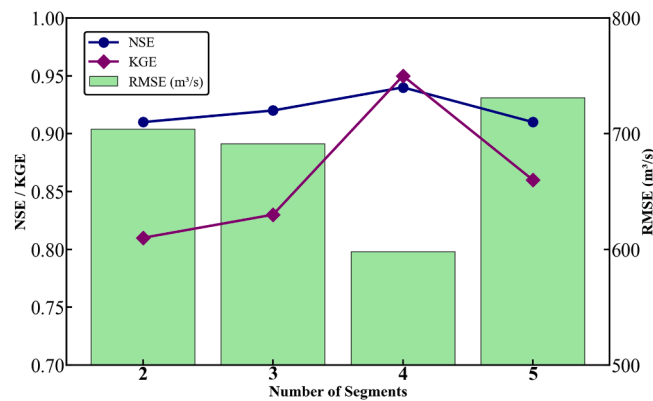


Fig. 7. Model performance under different segmentation schemes.

neural network weights to be jointly optimized. For clarity, the modeling schemes in this study are categorized into three groups: (1) Pure data-driven model — the Bayesian-optimized BO-BiLSTM; (2) Shallow physics-guided hybrid model — BO-Muskingum-BiLSTM (one-way coupling), in which the Muskingum component operates externally to the network; (3) Deeply coupled hybrid models — BO-Muskingum-BiLSTM (deep coupling), where the Muskingum routing process is embedded within the network and jointly optimized under two- to five-segment configurations. The specific input and output variables associated with each model are outlined in Table 4.

An analysis of the accuracy metrics presented in Table 5, alongside the scatter distribution patterns illustrated in Fig. 8, indicates that the structural complexity of models significantly influences the accuracy of runoff predictions. The pure machine learning model (S1) demonstrates satisfactory performance at low flow rates (below 5000 m³/s), yet exhibits considerable deviations in the medium to high flow range, characterized by a more dispersed point distribution. The NSE values for the training and testing periods are recorded at 0.91 and 0.90, respectively. As shown in Fig. 8, when flow rates exceed 5000 m³/s, the disparity between predicted and observed values becomes markedly pronounced, with certain high-flow samples deviating by more than 25 % from the 1:1 line. This observation highlights the limitations inherent in a pure machine learning model when attempting to capture complex hydrodynamic processes.

The S1 model exhibits overall performance comparable to the purely data-driven model (S0) under low and medium flow conditions, with a noticeable advantage in bias control. As shown in Table 5, it achieves NSE values of 0.92 and 0.90, RMSE values of 322 m³/s and 744 m³/s, and KGE values of 0.86 and 0.80 for the training and testing periods, respectively. Notably, although its NSE and RMSE are essentially equivalent to those of the pure BO-BiLSTM, the higher KGE values indicate that the one-way coupling introduces additional improvements in terms of correlation, bias, and variability structure. However, because this approach lacks joint optimization and feedback between the physical component and the neural network, its overall enhancement remains limited; certain static physical information may introduce redundancy or cumulative bias, thereby constraining further gains in predictive performance.

In contrast, the integration of physical mechanisms into hybrid model variants results in a nonmonotonic performance trend characterized by an initial improvement followed by a decline as the number of segments increases. For example, the two-segment BO-Muskingum-BiLSTM model achieves a reduction in test-period RMSE to 704 m³/s—representing a 6.6 % improvement over the pure machine learning model, while the scatter points increasingly align with the 1:1 line. The model attains optimal performance with four segments, yielding an NSE of 0.93 during training and 0.94 during testing, alongside a RMSE of 598 m³/s, which is 20.7 % lower than that of the pure machine learning model, and a KGE of 0.94 in training and 0.95 in testing. High-flow data points closely adhere to the identity line in Fig. 8, indicating that the four-segment configuration effectively captures the spatial heterogeneity of the channel.

It is noteworthy that increasing the segmentation to five segments results in a decline in test-period performance (KGE = 0.83), suggesting that excessive model complexity, or over-parameterization, can lead to unstable calibration and diminished generalization capabilities. Overall, the deeply integrated hybrid models (S2–S4) consistently outperform both the purely data-driven baseline model S0 and the one-way coupling model S1 across all flow regimes. The deeply integrated models also demonstrate superior goodness-of-fit across all flow regimes and significantly enhanced accuracy during high-flow events. This finding underscores the importance of incorporating physical process representations within machine learning frameworks to substantially improve generalization and

Table 4

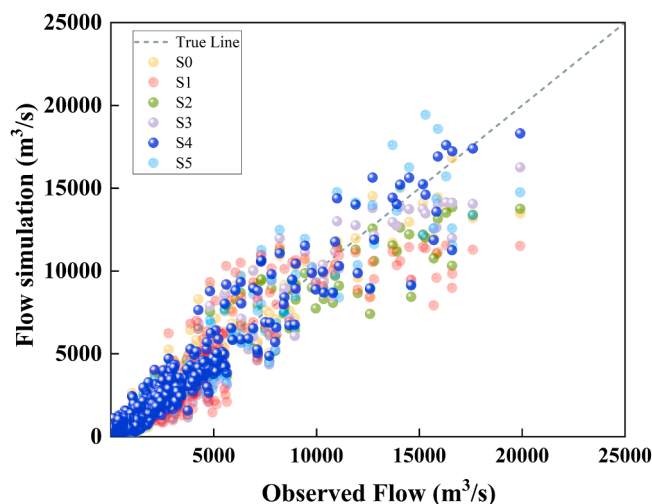
The input and output variables of each model.

Model	Number of river segments	Input variable	Output variable
S0 BO-BiLSTM	\	AK_O(t - 1), HLT_O(t - 1), E(t - 2), P(t - 2)	DJK_I(t)
S1 BO-Muskingum-BiLSTM (one-way coupling)	\		
S2 BO-Muskingum-BiLSTM (Deep coupling)	2		
S3	3		
S4	4		
S5	5		

Table 5

Training and testing accuracy metrics for different model schemes.

Model		Number of river segments	Training period			Test period		
			NSE	RMSE	KGE	NSE	RMSE	KGE
S0	BO-BiLSTM	\	0.91	352	0.85	0.90	754	0.76
S1	BO-Muskingum-BiLSTM(one-way coupling)	\	0.92	322	0.86	0.90	744	0.80
S2	BO-Muskingum-BiLSTM(Deep coupling)	2	0.92	337	0.86	0.91	704	0.81
S3		3	0.90	386	0.82	0.92	691	0.86
S4		4	0.93	316	0.94	0.94	598	0.95
S5		5	0.92	329	0.86	0.91	731	0.83

**Fig. 8.** Scatter of observed and simulated runoff for model schemes.

predictive accuracy. Among the models evaluated, the four-segment model S4 exhibited the best performance, highlighting the critical role of optimized reach segmentation in accurately capturing the spatial heterogeneity of river systems.

A comparative analysis was conducted between the pure machine learning model (S0), the one-way coupled physics-guided model (S1), and the deeply integrated hybrid model (S4), as shown in Fig. 9. The results indicate that the S4 model significantly surpasses both the S0 and S1 models in reproducing the shapes of flood peaks and maintaining phase synchronization. Notably, during extreme flood events (as shown in the inset), the hydrograph generated by the S4 model aligns closely with the observed data, whereas the S0 and S1 models considerably underestimate peak discharge and fail to capture the timing and intensity of these peaks. This comparison highlights the effectiveness of the S4 model in overcoming the limitations of purely data-driven and loosely coupled approaches under high-flow conditions. By explicitly capturing routing dynamics across spatially segmented river reaches, the S4 model provides substantial temporal support for decisions related to reservoir regulation and runoff forecasting.

3.4. Parameter sensitivity analysis

To quantify the influence of different parameters on model performance, a global variance-based Sobol sensitivity analysis was conducted. The Sobol method decomposes the total variance of the model output Y into contributions associated with individual parameters and their interactions (Sobol, 2001). For an input parameter vector $\theta = (\theta_1, \theta_2, \dots, \theta_M)$ and model output $Y = f(\theta)$ the total-effect index for parameter is θ_i is defined as:

$$S_{T,i} = 1 - \frac{\text{Var}_{\theta_{-i}}(E_0[Y | \theta_{-i}])}{\text{Var}(Y)} \quad (4)$$

Where $E_{\theta_i}[Y | \theta_{-i}]$ denotes the conditional expectation with respect to θ_i and $\text{Var}_{\theta_{-i}}(\sim)$ represents the remaining variance after excluding θ_i . Given that the total-effect Sobol index (S_T) provides an integrated measure of each parameter's overall influence, accounting for both its direct contribution and interactions with other parameters, it was adopted as the primary sensitivity metric to ensure robust evaluation under coupled parameter conditions.

The NSE and KGE were selected as the response variables in the Sobol analysis (Alipour et al., 2022; Buitink et al., 2020). NSE reflects the temporal dynamic agreement between simulated and observed flows, while KGE jointly assesses bias, correlation, and variability. Incorporating these metrics into the sensitivity framework allows parameter influences to be evaluated directly in terms of

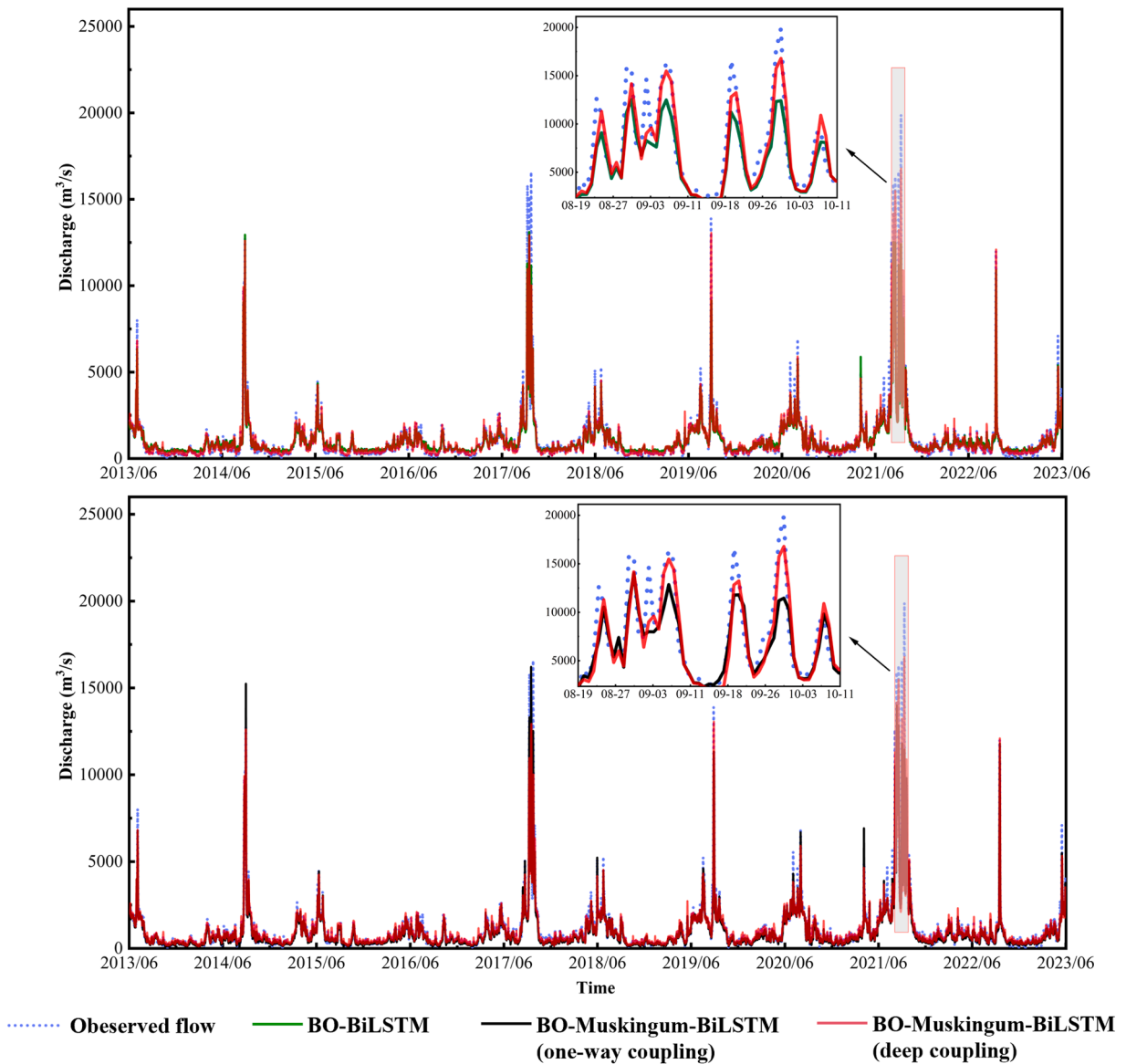


Fig. 9. Comparison of model-simulated runoff results.

predictive performance rather than through a single physical output variable. This enables a more comprehensive interpretation of how parameter perturbations affect hydrological accuracy.

A quasi-random Sobol sequence with $N = 256$ samples was used. For $M = 3$ parameters ($NumUnits$, K_1 , x_1), the total number of model evaluations followed the Sobol sampling formulation $N(M+2) = 1280$. The computed total-effect indices are summarized in Table 6.

The analysis was performed for the first segment of the differentiable routing layer within the optimal four-segment model, focusing on the storage coefficient (K_1), the inflow–outflow weighting factor (x_1), and the number of hidden units ($NumUnits$) in the neural network. The first-segment parameters were selected because they control the initial transformation of upstream inflow and exert the strongest upstream-to-downstream influence within the four-segment routing structure, thereby governing the propagation of flow dynamics throughout the subsequent reaches. The calculated total-effect indices are as follows: for the NSE metric, $S_T(K_1) = 0.657$, $S_T(x_1) = 0.815$, and $S_T(NumUnits) = 0.889$; for the KGE metric, $S_T(K_1) = 0.461$, $S_T(x_1) = 0.504$, and $S_T(NumUnits) = 0.920$. These results indicate that $NumUnits$ consistently shows the largest contribution to model performance variance, suggesting that the representation capacity of the neural component is the most influential factor in determining predictive stability and accuracy.

The weighting factor x_1 also exhibits a strong impact, implying that the inflow–outflow balance within the differentiable routing structure plays an essential role in reproducing dynamic responses. By contrast, the storage coefficient K_1 has a relatively lower total effect, indicating that its influence becomes secondary once the flow routing characteristics are well captured. Overall, the results

Table 6
Sobol total-effect indices of parameters.

Parameter	NumUnits	K_1	x_1
Sobol index S_T for NSE	0.889	0.657	0.815
Sobol index S_T for NSE	0.920	0.461	0.504

reveal a hierarchical sensitivity structure rather than shallow joint control: the neural architecture parameter (*NumUnits*) dominates, the physical weighting parameter (x_1) provides key physical constraints, and the storage coefficient (K_1) acts as a fine-tuning factor affecting response timing.

3.5. Regional hydrological implications and limitations

The present study illustrates a general paradigm of physically coupled machine learning, where hydrological process modules are integrated with data-driven architectures to achieve a balance between theoretical interpretability and predictive flexibility. As exemplified by the differentiable Muskingum-BiLSTM framework developed in this study, the model achieves end-to-end learning by embedding a physically parameterized routing process within the neural network, such approaches provide a transferable foundation for hybrid hydrological modeling. Similar ideas have also been discussed by (Tian et al., 2025), who developed an ensemble calibration framework for flood forecasting based on the response curve of a rainfall dynamic system and LSTM. Their results demonstrate that coupling physical mechanisms with neural models can enhance both stability and accuracy in hydrological forecasting, aligning well with the methodology and philosophy of this study. Beyond reservoir inflow forecasting, the proposed paradigm can be extended to sediment transport, water–energy coupling, and eco-hydrological simulations, where embedding physical processes within machine learning structures may improve model robustness and generalization under nonstationary conditions. This direction represents a broader methodological pathway for unifying process-based understanding and data-driven intelligence in hydrological science.

From a regional hydrological perspective, it is essential to capture how human regulation propagates spatially across a basin. This process requires models that balance physical consistency with dynamic adaptability. The differentiable modeling structure introduced in this study allows the model to respond flexibly to diverse hydrological characteristics in different regions. In the upper reaches of the Danjiangkou Reservoir basin, the joint regulatory effects of the mainstem and tributary reservoirs create pronounced along-channel variability, which the segmented structure adopted in this study is particularly effective at capturing. As a result, the model achieves more accurate and physically coherent representations of spatially coupled processes within cascade systems. It should be noted that the model's performance partly depends on the accuracy of upstream reservoir discharge data. Due to the nonlinear nature of runoff processes, even small upstream errors may propagate downstream, affecting parameter calibration and inflow prediction. Moreover, the generalization ability is constrained by the climatic representativeness of training data—insufficient coverage of long-term variability may reduce performance under nonstationary conditions. Although the framework performs well for regulated systems, its applicability to natural rivers remains limited, as irregular flow processes disrupt the continuity of storage–discharge relationships. Nevertheless, the differentiable design enhances the framework's adaptability and provides a foundation for extending it to unregulated systems. Future research will focus on improving the model's adaptability to unregulated river systems and enhancing its robustness under diverse climatic and hydrological conditions.

4. Conclusion

This study explored the feasibility of introducing differentiable physical mechanisms into river routing simulation, combining physical constraints with data-driven learning. The results demonstrate that this approach effectively reproduces the dynamic responses of cascade reservoir systems and significantly improves the accuracy of flood peak timing and magnitude. The key contributions of this study include:

(1) The integration of a differentiable Muskingum layer with BiLSTM forms a coupled model that retains both interpretability and the theoretical foundation of the Muskingum method. Meanwhile, the adaptive capabilities of deep learning allow the model to capture complex hydrological dynamics more effectively, thereby overcoming the limitations of traditional physical models and improving prediction accuracy.

(2) By applying BO to optimize both the segmented Muskingum parameters and the neural network architecture, the model effectively addresses the challenges of parameter calibration, resulting in enhanced performance and robustness.

(3) Segmenting the river into sub-reaches with tailored Muskingum parameters helps represent spatial variability in hydraulic behavior. This segmentation leads to a more accurate description of runoff propagation across the river system. Empirical results for the Ankang–Danjiangkou reach show that the four-segment hybrid model achieved a NSE of 0.94, a KGE of 0.95, and an RMSE of 598 m³/s during the test period. Compared with both the pure BiLSTM model and the one-way coupled physics-guided model, the hybrid approach showed significantly better alignment with observed peak flows and more concentrated scatter near the 1:1 line. This approach significantly enhances the model's ability to capture flood peak timing and magnitude, thereby improving the representation of reservoir-river system responses.

CRediT authorship contribution statement

Donglin Gu: Writing – review & editing, Data curation. **Jianbo Chang:** Writing – review & editing, Visualization, Conceptualization. **Dongxu Yang:** Writing – original draft, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Baowei Yan:** Writing – review & editing, Validation, Supervision, Resources, Funding acquisition, Conceptualization. **Shixiong Du:** Writing – review & editing, Visualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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